

Data Cleaning & Reporting Automation

Automated preprocessing, quality control, reporting, and visual summary pipeline

Dataset note: This project uses a synthetic/illustrative dataset created specifically for demonstrating data-cleaning automation. It is not real company data.

1. Objective

Build a reproducible Python workflow that detects and handles common data-quality problems, produces cleaned data, and automatically generates summary reports and visualizations.

2. Data-quality issues addressed

- Missing values in region, category, quantity, unit price, and revenue fields.
- Exact duplicate rows.
- Inconsistent capitalization and whitespace in product, category, and region values.
- Mixed date formats and invalid dates.
- Currency symbols and thousands separators stored inside numeric fields.
- Invalid quantity/price values.
- Revenue values inconsistent with quantity multiplied by unit price.

3. Cleaning methodology

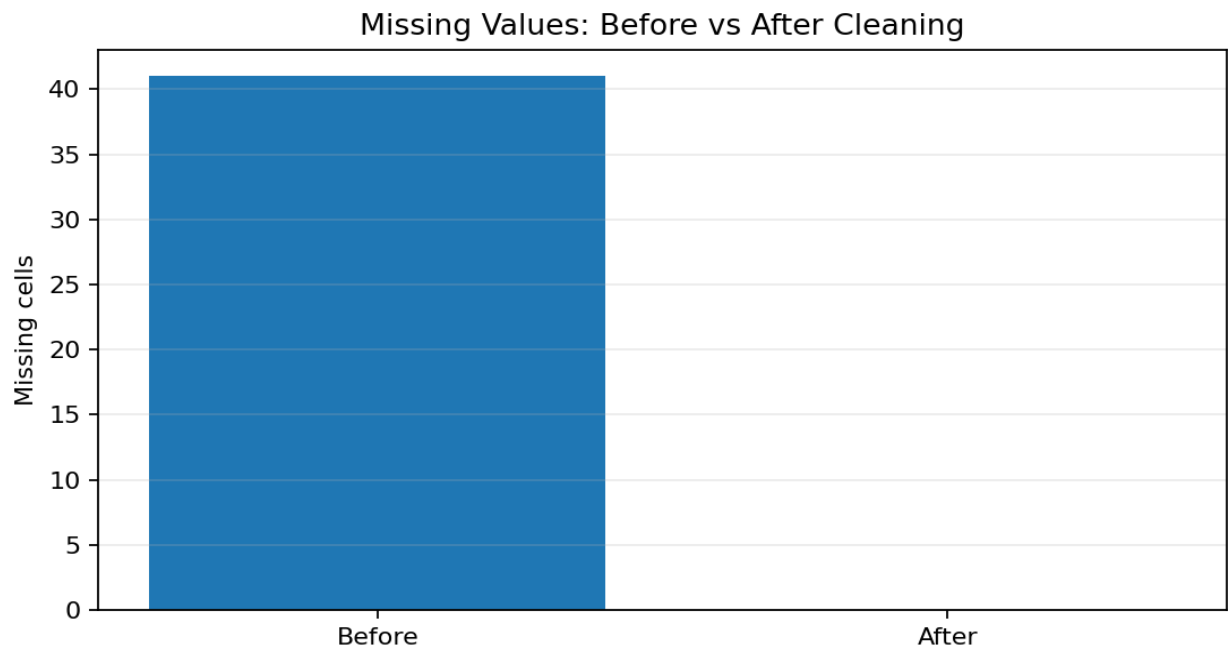
- Strip whitespace and standardize known product, category, and region aliases.
- Parse mixed date formats and remove records with invalid essential dates.
- Convert quantities and prices to numeric values; remove non-positive quantities/prices.
- Impute missing quantity using the median quantity for the standardized product.
- Impute missing unit price using the median unit price for the standardized product, with a global median fallback.
- Fill missing region/category values with documented fallback rules where possible.
- Recalculate revenue as Quantity x Unit_Price to correct missing or inconsistent revenue values.
- Remove duplicate records before and after standardization.

4. Cleaning results

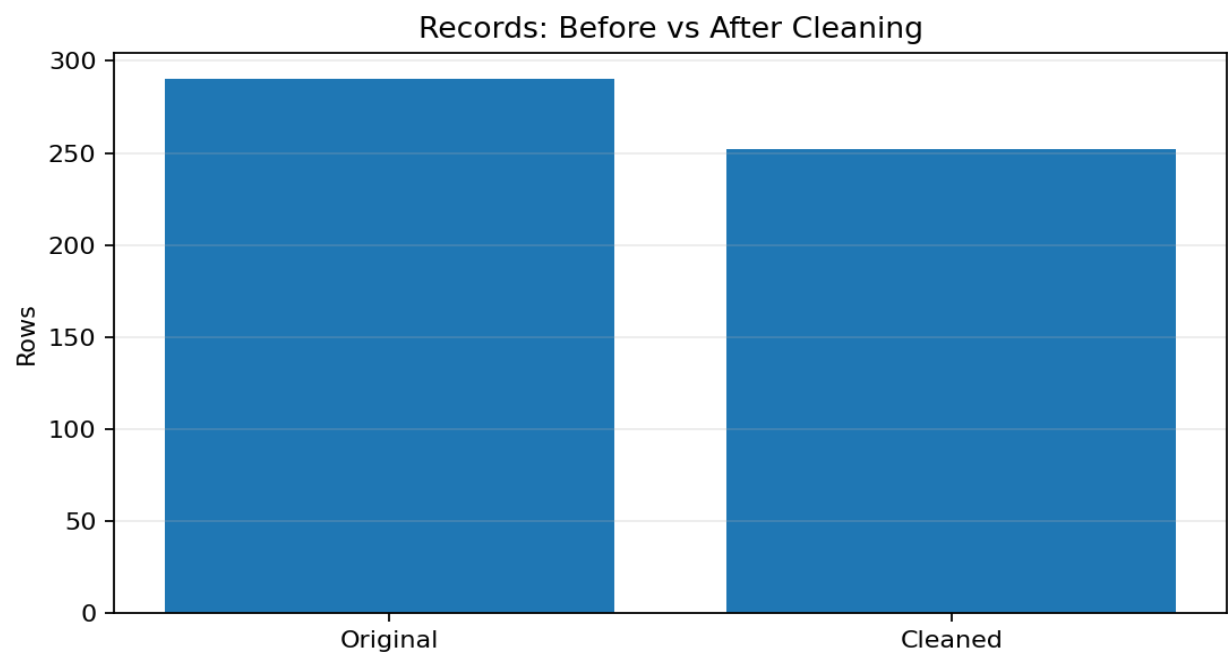
Metric	Value
Original rows	290
Original columns	8
Duplicate rows removed	10
Invalid/removed rows	9
Final rows	252
Missing cells before	41
Missing cells after	0
Standardized text changes	127
Revenue corrections	22

5. Automated visual summaries

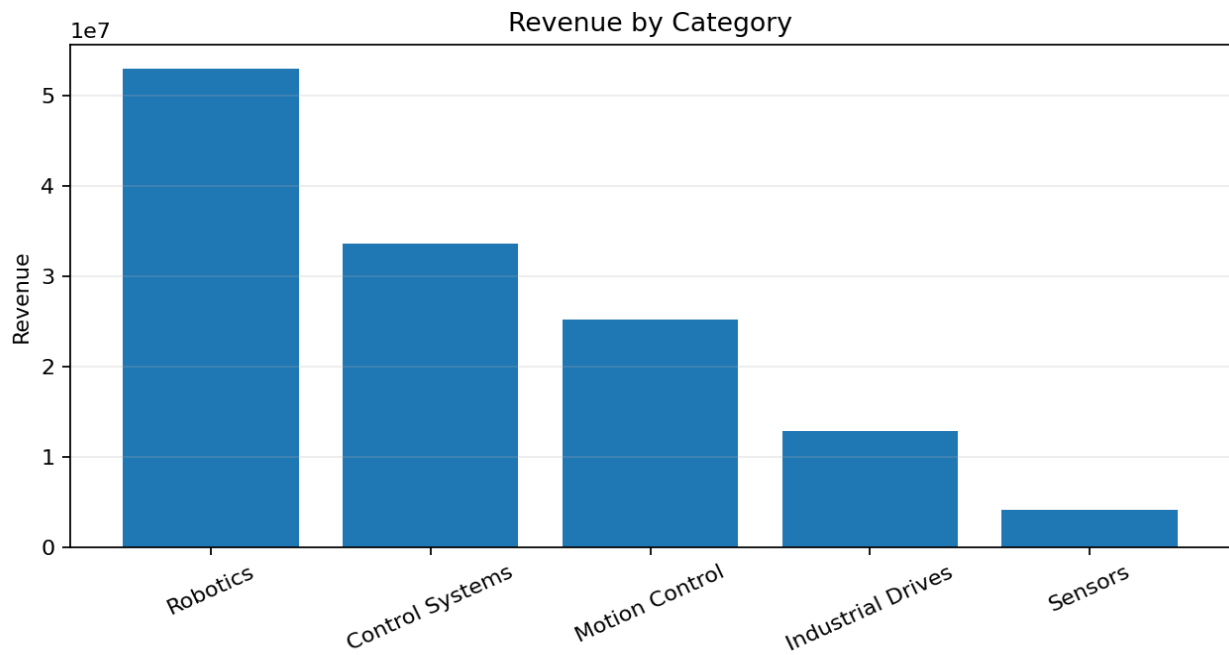
Missing values before vs after cleaning



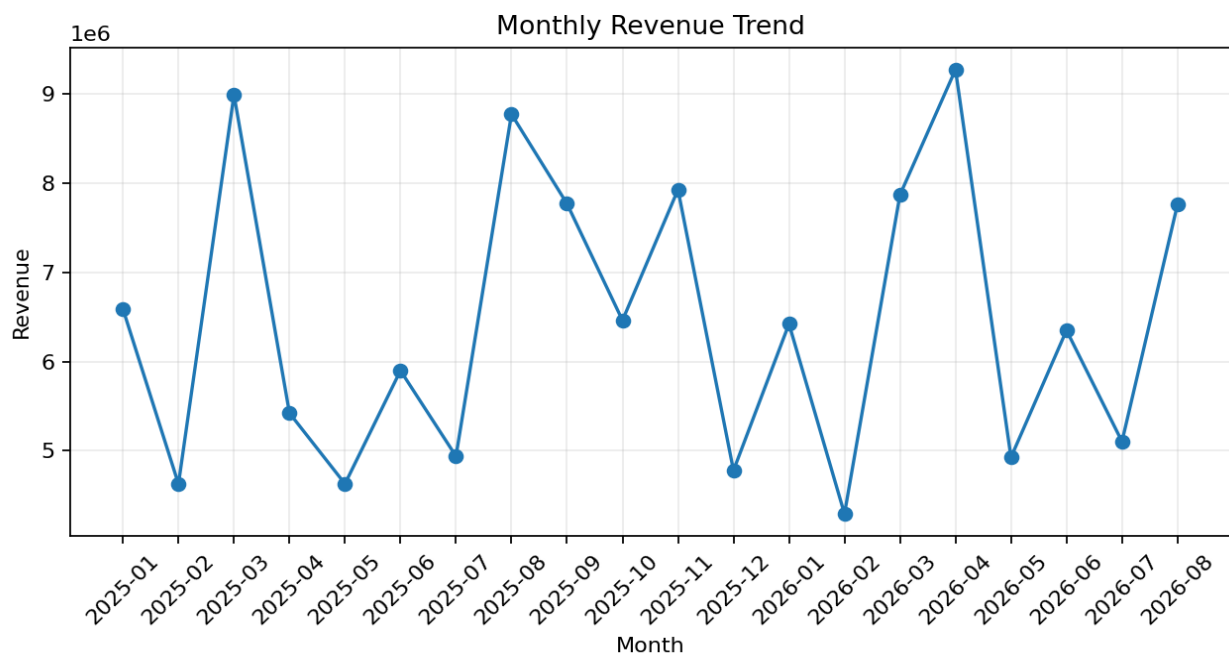
Records before vs after cleaning



Category distribution



Monthly trend



6. Summary outputs

The workflow saves machine-readable outputs for further analysis and reproducibility: `cleaned_data.csv`, `cleaning_summary.csv`, `summary_statistics.csv`, `category_summary.csv`, `monthly_summary.csv`, and `automated_report.txt`.

7. Conclusion

The completed pipeline turns intentionally messy source data into a standardized dataset while documenting what was changed. The same script can be rerun on a compatible CSV or Excel input, making the cleaning and reporting process repeatable and efficient.

8. Future improvements

Possible extensions include configurable validation rules, automated email delivery of reports, scheduled execution, anomaly-detection checks, interactive dashboards, and database connectivity.